

LESSON 5-8 • SESSION 1 • PRACTICE • FORM A

Surface Area of Pyramids

Learning target – I can find the surface area of a pyramid by adding the base area and the areas of its triangular faces.
6.G.A.4

REMEMBER

- **SA of a pyramid** = base area + area of all the **triangular faces**.
- Each triangle: $\frac{1}{2} \times$ base edge $\times$ **slant height** (not the straight-up height).
- A square pyramid has 1 square + **4** matching triangles.

WORKED EXAMPLE

Square base 10 cm; slant height 7 cm.

Step 1 Base: $10 \times 10 = 100 \text{ cm}^2$.

Step 2 One triangle: $\frac{1}{2} \times 10 \times 7 = 35 \text{ cm}^2$.

Step 3 Four triangles: 140. Add: $100 + 140$.

Answer SA = 240 cm²

WORDS **slant height** (*altura inclinada*) • **lateral face** (*cara lateral*) • **apex** (*ápice*)

PRACTICE

show your work • label units

1. Find the surface area of each square pyramid.

a. base 8 m, slant height 10 m ____ m²

b. base 5 ft, slant height 8 ft ____ ft²

c. base 2 cm, slant height 3 cm ____ cm²

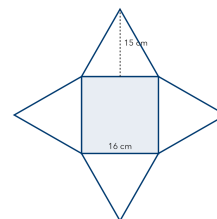
d. base 7 in, slant height 10 in ____ in²

2. **Pyramid Model.** A class model is a square pyramid. Its net is shown.

a. Area of the square base ____ cm²

b. Area of one triangle ____ cm²

c. Surface area ____ cm²



3. **Gift Topper.** A gift topper is a square pyramid with a base 8 in on each side and a slant height of 11 in. Find its surface area.

Hint Base first, then four triangles.

Show your work.

Answer _____ in²

4. **Trophy Top.** A trophy top is a square pyramid with a base 12 cm on each side and a slant height of 9 cm. Every face is plated. Find its surface area.

Show your work.

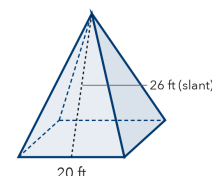
Answer _____ cm²

5. **Roof Cap.** Every face of this roof cap, including the bottom, will be shingled. Find its surface area.

Hint Use the slant height for each triangle.

Show your work.

Answer _____ ft²



6. **Priya's Pyramid.** A square pyramid has base 18 cm and slant height 13 cm. Priya says its surface area is 117 cm^2 . Is Priya right?

Hint Which faces of the net are missing from 117?

Show your work.

Answer _____ cm^2

7. **Sand Mold.** A sand mold is a square pyramid: base 16 in, slant height 10 in. Every face counts. Find its surface area.

Hint Add the base to all 4 triangles.

Show your work.

Answer _____ in^2

8. **Museum Model.** A museum model is a square pyramid: base 14 in, slant height 15 in. Gold foil covers only the four triangles. Foil costs \$0.04 per in^2 .

- a. Base area and one triangle _____
 b. Total surface area _____ in^2
 c. Cost of the foil _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Ethan's Work.** A square pyramid has a base 22 in on each side and a slant height of 17 in. Ethan wrote: " $\text{SA} = 484 + 187 = 671 \text{ in}^2$." Explain the mistake and fix it.

TEST PRACTICE

choose the best answer

10. How many triangular faces does a square pyramid have?

A 3 **B** 4 **C** 5 **D** 6

11. A base of 25 cm^2 plus four triangles of 20 cm^2 each gives a surface area of –

A 45 cm^2 **B** 80 cm^2 **C** 100 cm^2 **D** 105 cm^2

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

360 R	576 I	16.80 S	1,040 Z	240 Y	736 P	1,440 A	616 D	792 M	117 Q
#2c	#3	#4	#5	#6	#7	#8b	#8c		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____